

SIMATS Online Programming Lab

Java Programming — CO4-AT2-Laboratory Performance
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Q3. Student Record

Marks: 25 marks

Question: Design a GUI application that allows the user to maintain a simple student record (add/update/delete/display) using appropriate layout manager, controls, and event handling.

Source Code

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.*;
import javax.imageio.ImageIO;
import java.io.File;

public class StudentRecordApp extends JFrame implements ActionListener {

    JTextField idField, nameField, markField;
    JTable table;
    DefaultTableModel model;
    JButton addBtn, updateBtn, deleteBtn, displayBtn;

    public StudentRecordApp() {
        setTitle("Kumaran S (192372048RB) - Student Record Manager");
        setSize(560, 420);
        setLayout(new BorderLayout());

        JLabel nameLabel = new JLabel("Kumaran S - 192372048RB", SwingConstants.CENTER);
        nameLabel.setFont(new Font("SansSerif", Font.BOLD, 16));
        nameLabel.setOpaque(true);
        nameLabel.setBackground(new Color(25, 55, 110));
        nameLabel.setForeground(Color.WHITE);
        nameLabel.setBorder(BorderFactory.createEmptyBorder(8, 8, 8, 8));

        JPanel formPanel = new JPanel(new GridLayout(1, 6, 5, 5));
        formPanel.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));
        formPanel.add(new JLabel("ID:"));
        idField = new JTextField();
        formPanel.add(idField);
        formPanel.add(new JLabel("Name:"));
        nameField = new JTextField();
        formPanel.add(nameField);
        formPanel.add(new JLabel("Mark:"));
        markField = new JTextField();
        formPanel.add(markField);

        JPanel topPanel = new JPanel(new BorderLayout());
        topPanel.add(nameLabel, BorderLayout.NORTH);
        topPanel.add(formPanel, BorderLayout.SOUTH);
        add(topPanel, BorderLayout.NORTH);

        model = new DefaultTableModel(new Object[]{"ID", "Name", "Mark"}, 0);
        table = new JTable(model);
        add(new JScrollPane(table), BorderLayout.CENTER);

        JPanel btnPanel = new JPanel(new FlowLayout());
        addBtn = new JButton("Add");
        updateBtn = new JButton("Update");
        deleteBtn = new JButton("Delete");
        displayBtn = new JButton("Display All");
        btnPanel.add(addBtn);
        btnPanel.add(updateBtn);
        btnPanel.add(deleteBtn);
        btnPanel.add(displayBtn);
        add(btnPanel, BorderLayout.SOUTH);
        addBtn.addActionListener(this);
```

```

        updateBtn.addActionListener(this);
        deleteBtn.addActionListener(this);
        displayBtn.addActionListener(this);

        table.getSelectionModel().addListSelectionListener(e -> {
            int row = table.getSelectedRow();
            if (row >= 0) {
                idField.setText(model.getValueAt(row, 0).toString());
                nameField.setText(model.getValueAt(row, 1).toString());
                markField.setText(model.getValueAt(row, 2).toString());
            }
        });

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        Object src = e.getSource();
        if (src == addBtn) {
            model.addRow(new Object[]{idField.getText(), nameField.getText(), markField.getText()});
        } else if (src == updateBtn) {
            int row = table.getSelectedRow();
            if (row >= 0) {
                model.setValueAt(idField.getText(), row, 0);
                model.setValueAt(nameField.getText(), row, 1);
                model.setValueAt(markField.getText(), row, 2);
            }
        } else if (src == deleteBtn) {
            int row = table.getSelectedRow();
            if (row >= 0) model.removeRow(row);
        } else if (src == displayBtn) {
            StringBuilder sb = new StringBuilder("ID\tName\tMark\n");
            for (int i = 0; i < model.getRowCount(); i++) {
                sb.append(model.getValueAt(i, 0)).append("\t")
                  .append(model.getValueAt(i, 1)).append("\t")
                  .append(model.getValueAt(i, 2)).append("\n");
            }
            JOptionPane.showMessageDialog(this, sb.toString(), "All Student Records", JOptionPane.INFORMATION_MESSAGE);
        }
    }

    private void captureScreenshot(String path) {
        try {
            Rectangle rect = this.getBounds();
            Robot robot = new Robot();
            Thread.sleep(300);
            java.awt.image.BufferedImage img = robot.createScreenCapture(rect);
            ImageIO.write(img, "png", new File(path));
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static void main(String[] args) throws Exception {
        StudentRecordApp app = new StudentRecordApp();
        SwingUtilities.invokeLater(() -> {
            app.idField.setText("101"); app.nameField.setText("Arun"); app.markField.setText("88");
            app.addBtn.doClick();
            app.idField.setText("102"); app.nameField.setText("Kumaran"); app.markField.setText("92");
            app.addBtn.doClick();
            app.idField.setText("103"); app.nameField.setText("Priya"); app.markField.setText("79");
            app.addBtn.doClick();
        });
        Thread.sleep(700);
        app.captureScreenshot("/home/claude/lab/screenshots/q3_output.png");
        System.exit(0);
    }
}

```

Output Screenshot

Kumaran S - 192372048RB

ID: Name: Mark:

ID	Name	Mark
101	Arun	88
102	Kumaran	92
103	Priya	79

Add

Update

Delete

Display All